

React JS



Agenda

- Custom Hook
- React-query
- Recap
- General Topics
- General Resources
- Questions

Custom hook

Hazem Alsaqaan

```
1 import axios from "axios";
2 import { useEffect, useState } from "react";
3
4 const useGetData = (url) => {
5   const [data, setData] = useState([]);
6   const [loading, setLoading] = useState(true);
7   const [error, setError] = useState(null);
8
9   useEffect(() => {
10     const getProducts = async () => {
11       try {
12         setLoading(true);
13         const response = await axios.get(url);
14         setData(response.data.products);
15       } catch (err) {
16         setError(err);
17       } finally {
18         setLoading(false);
19       }
20     };
21
22     getProducts();
23   }, [url]);
24
25   return { data, setData, loading, error };
26 };
27
28 export default useGetData;
```

React Query

[tanstack/react-query](https://tanstack.com/react-query)

React Query : install



Hazem Alsaqaan

1

2 `npm install @tanstack/react-query`

React Query : provider

Hazem Alsaqaan



```
1  import { QueryClient, QueryClientProvider } from "@tanstack/react-query";
2
3  const queryClient = new QueryClient();
4  createRoot(document.getElementById("root")).render(
5    <QueryClientProvider client={queryClient}>
6      <App />
7    </QueryClientProvider>,
8  );
9
```

React Query : fetch data

Hazem Alsaqaan

```
1  import { useQuery } from "@tanstack/react-query";
2  import axios from "axios";
3
4  const getProducts = async () => {
5    const res = await axios.get(`https://dummyjson.com/products`);
6    return res.data;
7  };
8
9  const useGetData = () => {
10    const query = useQuery({
11      queryKey: ["products"],
12      queryFn: getProducts,
13    });
14    return query;
15  };
16
17  export default useGetData;
```

React Query : Use in component

Hazem Alsaqaan



```
1  function ProductsPage() {
2    const { data, isLoading, isError, error } = useGetData();
3    if (isLoading) {
4      return <div>loading...</div>;
5    }
6    if (isError) {
7      return <div>{error.message}</div>;
8    }
9    return(
10     <>{data.products.map((item)=>(<p key={item.id}>{item.title}</p>))}</>
11   )
12 }
13 export default ProductsPage;
```


React-Query

Dev Tool

React Query : Dev-Tool

Install



```
1 // npm i @tanstack/react-query-devtools
```

Hazem Alsaqaan

Import



```
1 import { ReactQueryDevtools } from "@tanstack/react-query-devtools";  
2  
3 createRoot(document.getElementById("root")).render(  
4   <BrowserRouter>  
5     <QueryClientProvider client={queryClient}>  
6       <App />  
7       <ReactQueryDevtools />  
8     </QueryClientProvider>  
9   </BrowserRouter>  
10 );  
11
```

Hazem Alsaqaan

Resources

For All

Resources

- **Crash Course:**

- <https://www.youtube.com/watch?v=4baq00tHfmA&t=24s>
- <https://www.youtube.com/watch?v=u6gSSpfsoOQ&t=123s>
- <https://www.youtube.com/watch?v=SqcY0GIETPk>
- <https://www.youtube.com/watch?v=2-crBg6wpp0>
- <https://www.youtube.com/watch?v=dCLhUialKPQ>
- https://www.youtube.com/watch?v=Bvwq_S0n2pk

- **Interactive Learning Course:**

- <https://scrimba.com/learn/learnreact>

- **React Hooks**

- <https://www.youtube.com/watch?v=OriHps91AzE>
- <https://www.youtube.com/watch?v=TNhaISOuy6Q&t=26s>
- <https://www.youtube.com/playlist?list=PLtxOBbrOOPH4ro6EXTNHrlvmoNaOcPAwe>

- **Forms in deep**

- <https://www.youtube.com/watch?v=a94FOvaBomQ&list=PLC3y8-rFHvwiPmFbtzEWjESkqBVDbdgGu>
- <https://www.youtube.com/watch?v=KejZXxFCe2k&list=PLC3y8-rFHvwjmgBr1327BA5bVXoQH-w5s>

Resources

- **Redux**

- <https://www.youtube.com/watch?v=u3KlatzB7GM&list=PL0Zuz27SZ-6M1J5I1w2-uZx36Qp6qhjKo>
- https://www.youtube.com/watch?v=_shA5Xwe8_4&t=1s

- **Chat with firebase**

- <https://www.youtube.com/watch?v=zQyrwxMPm88>

- **React**

- https://www.youtube.com/watch?v=N0DhCV_-Qbg
- <https://www.youtube.com/watch?v=uybSanJSQng>
- <https://react.dev/blog/2024/12/05/react-19>

- **React router V6**

- <https://www.youtube.com/watch?v=zEQiNFAwDGo&t=84s>
- <https://www.youtube.com/watch?v=L2kzUg6lzxM&t=136s>
- <https://scrimba.com/learn/reactrouter6>

Resources

- **Translation (Format.js)**
 - <https://formatjs.io/>
 - <https://blog.logrocket.com/translate-react-app-using-format-js/>
- **Redux vs Context**
 - <https://betterprogramming.pub/my-frustrations-with-the-context-api-in-react-26189fcd5371>

Thank you

React JS



Agenda

- **Forms**
- **Protected Route**
- **Code Splitting**
- **General Topics**
- **Open discussion and questions**
- **Recap**

Forms

Forms in React

We can handle forms in different ways:

Here's the most native and simple ways to handle it using hooks,

<https://www.freecodecamp.org/news/how-to-build-forms-in-react/>

With React 19 new hooks appeared to handle the forms on servers directly (**self study**)

server-side functions you can call directly from your React components (or form submissions) without building a separate API layer.

<https://www.youtube.com/watch?v=ExZUdkfu-KE&t=443s>

Forms in React

For More Organized, complex and advanced forms handling, You can use forms third parties like

- Formik with Yup for validation

<https://formik.org/>

- React hook form with Zod

<https://react-hook-form.com/>

Code Splitting

Code Splitting

Bundling is great, but as your app grows, your bundle will grow too. Especially if you are including large third-party libraries. You need to keep an eye on the code you are including in your bundle so that you don't accidentally make it so large that your app takes a long time to load.

To avoid winding up with a large bundle, it's good to get ahead of the problem and start "splitting" your bundle

<https://dev.to/franklin030601/code-splitting-in-react-js-4o2g>

Code Splitting

import()

The best way to introduce code-splitting into your app is through the dynamic `import()` syntax.

React.lazy

The `React.lazy` function lets you render a dynamic import as a regular component.

Code Splitting

Before:

```
import OtherComponent from './OtherComponent';
```

After:

```
const OtherComponent = React.lazy(() => import('./OtherComponent'));
```

This will automatically load the bundle containing the OtherComponent when this component is first rendered. React.lazy takes a function that must call a dynamic import(). This must return a Promise which resolves to a module with a default export containing a React component.

Code Splitting

The lazy component should then be rendered inside a `Suspense` component, which allows us to show some fallback content (such as a loading indicator) while we're waiting for the lazy component to load.

Example:

```
import { Suspense } from 'react';
```

```
<Suspense fallback={<div>Loading...</div>}>
```

```
  <OtherComponent />
```

```
</Suspense>
```

General Topics

Env variables

<https://vite.dev/guide/env-and-mode>

- Create .env file at the same level of src folder
- Any environment variable should start with VITE then the key name like
 - VITE_APP_BASE_URL=<http://test.com>
- You can use the env variable in you code using the following
 - `{import.meta.env.VITE_APP_BASE_URL}`

Thank you

lap

Task : Register

In the register page with the following fields. [Native]

- Email address [required—email format] (regex)
- Name [required]
- Username [required—contains no spaces]
- Password [required—password length not less than 8 characters, contains at least one uppercase letter]
- Confirm password: [required—matches previous password]
- **[Bonus]** Try to show all errors when the user clicks on the register button at the beginning.
- After the user submits the form, redirect the user to the home page (products list)

The image shows a registration form with the following fields and a button:

- Name**: A text input field.
- Email**: A text input field.
- User Name**: A text input field.
- Password**: A text input field.
- Confirm Password**: A text input field.
- Register**: A green button with white text.

Task : Contact us

In the contact us page apply the following fields with validation

- Email address [required—email format]
- First Name [required]
- Last Name [required]
- Phone Number [optional]
- Message [Required, min 10 characters , max 500]

After submit , show a success message that we will get to you soon, then reset the form values.

- **[Bonus]** Extra, try to use a phone number package with countries and phone number formats

Send Us a Message

Please fill in the form below to get in touch with us.

☐ I've read and agree with [Terms of Service](#) and [Privacy Policy](#)